

Hotel Booking Analysis Project

Project Overview

This project analyzes hotel booking data to understand booking behavior, revenue performance, cancellation patterns, and customer segments.

The project was developed using Power BI and includes data cleaning, data modeling, KPI development, interactive dashboards, and business insights generation.

Objectives

- Analyze overall booking performance.
 - Identify revenue drivers and top-performing markets.
 - Investigate cancellation behavior and its impact on revenue.
 - Understand customer booking patterns.
 - Support business decision-making through interactive dashboards.
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Tools & Technologies

- Power BI
 - Power Query
 - DAX
 - Data Modeling
 - Star Schema
 - Data Visualization
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Data Cleaning and Data Quality Assessment

Before starting the data modeling, business analysis, and dashboard development phases, a comprehensive data quality assessment was conducted to validate data consistency, identify potential anomalies, and ensure the dataset was suitable for analytical purposes.

1. Missing Values Assessment

A complete review of all columns was performed to identify missing values.

Findings

Several columns contained missing values.

- The Children column contained 4 records with the value NA.
- Additional missing values were identified in fields such as Agent, Company, and a small number of records in Country.
- Missing values in Agent and Company are considered valid business cases, as many bookings were made without an associated travel agency or company.

Action Taken

- NA values in the Children column were converted to Null values.
- Missing values in Agent and Company were retained because they represent legitimate booking scenarios rather than data quality issues.
- Records containing missing Children values were retained due to insufficient information to infer the correct number of children.

2. Data Type Validation

All columns were reviewed to ensure that each field was assigned the appropriate data type.

Actions Taken

- Numeric fields were converted to numerical data types where required.
- Date fields were converted to proper date formats.
- Categorical fields were validated and retained as text attributes.

No critical data type issues remained after validation.

3. Creation of Derived Analytical Fields

To support business analysis and KPI calculations, additional analytical fields were created.

Total Guests

Total Guests = Adults + Children + Babies

Total Nights

Total Nights = Stays in Weekend Nights + Stays in Week Nights

Booking Date

A booking date field was derived using the reservation arrival date and lead time information to support booking behavior analysis and trend reporting.

4. Investigation of Missing Guest Counts

Findings

- 4 records produced Null values.
- These records correspond exactly to the observations where the Children field contained missing values.

Action Taken

- The records were retained.
- No assumptions were made regarding the missing number of children.

5. Guest Count Anomalies

Conditions Examined

- Total Guests = 0
- Reservation not canceled
- Positive stay duration
- Positive ADR values

Findings

- 31 records met these conditions.

Action Taken

- The records were retained due to their extremely small proportion within the dataset.
- The anomaly was documented for transparency.

6. ADR Anomalies

Findings

- 492 records had ADR equal to zero.
- These reservations were not canceled.
- They contained valid guest counts.

- They also contained valid stay durations.

Action Taken

- The records were retained.
- The anomaly was documented for analytical transparency.

7. Zero-Night Reservations

Findings

- 680 records had Total Nights equal to zero.
- All of these records also had ADR equal to zero.
- Reservation status indicated successful checkout rather than cancellation.

Action Taken

- The records were retained.
- No filtering was applied due to the absence of sufficient business evidence.

8. Duplicate Records Assessment

Findings

A substantial number of fully matching rows were identified.

However:

- The dataset does not contain a unique Booking ID.
- It is impossible to determine whether these observations are true duplicate records or separate reservations sharing identical attributes.

Action Taken

- No records were removed.
- Potential duplicates were retained to avoid accidental loss of valid reservations.

9. Geographic Data Enhancement

Actions Taken

A separate geographic lookup table was introduced containing:

- Country Code

- Country Name
- Continent

The lookup table was linked to the fact table through the country code field.

Additionally:

- The country code TMP was standardized to TLS to ensure successful matching with the geographic reference table.

10. Date Modeling Preparation

The original arrival date components were combined into a single Date field.

A dedicated Date Dimension table was created containing:

- Year
- Quarter
- Month
- Month Number
- Year-Month
- Weekday Name
- Weekday Number
- Weekend Indicator

11. Customer Segmentation Preparation

A customer classification field was created based on the original repeated guest indicator.

Guest Type

- New Guest
- Returning Guest

12. Final Data Quality Status

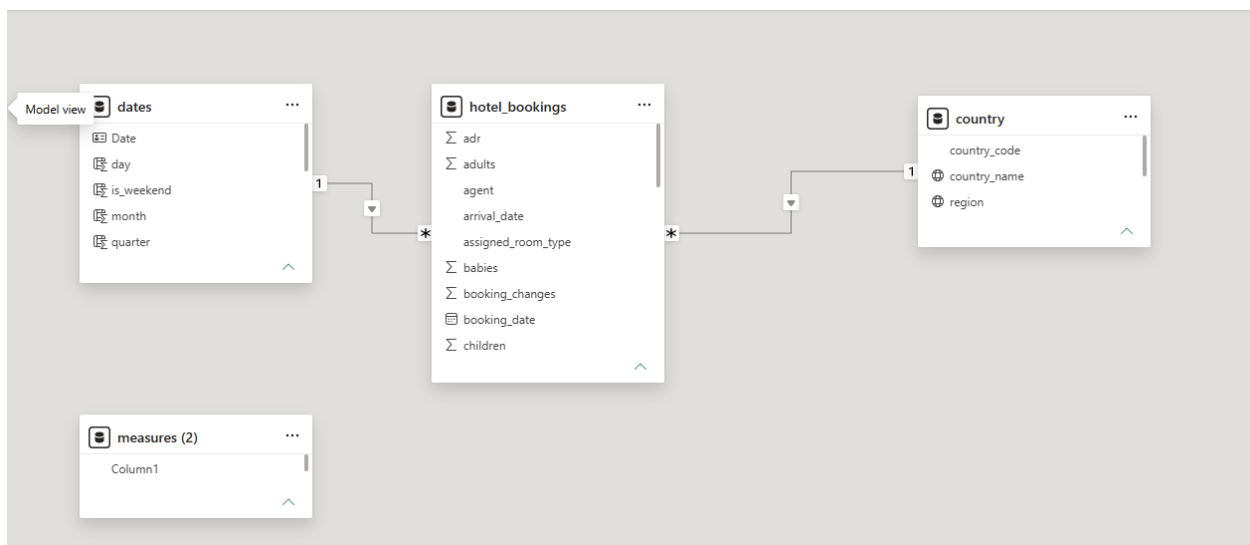
After completing the data preparation process:

- Data types were validated and corrected.
- Missing values were identified and investigated.

- Potential anomalies were documented.
- Derived analytical fields were created.
- Geographic enrichment was applied.
- Date modeling structures were established.
- Customer segmentation fields were prepared.
- No critical data quality issues were identified that justified removing a significant portion of the dataset.

The dataset was therefore considered suitable for data modeling, business analysis, KPI development, and dashboard construction.

Data Model



Dashboard Structure

Cover Page

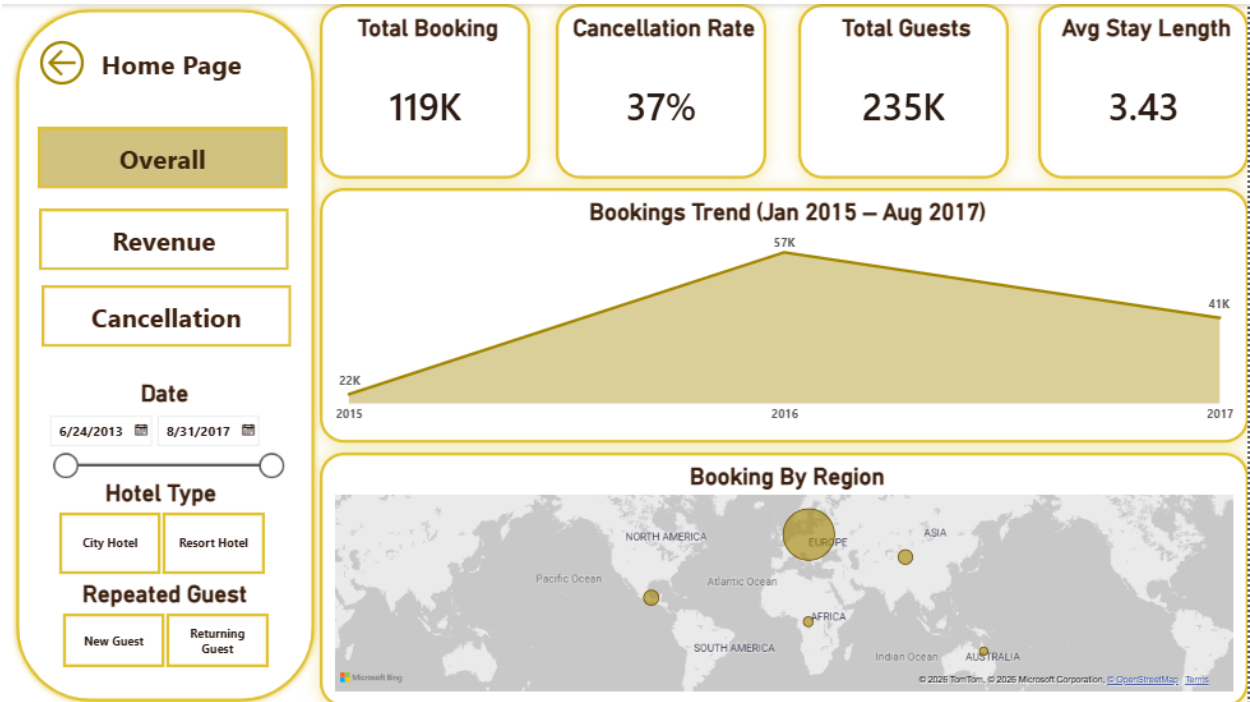
Project introduction and navigation.



Overall Performance Dashboard

Includes:

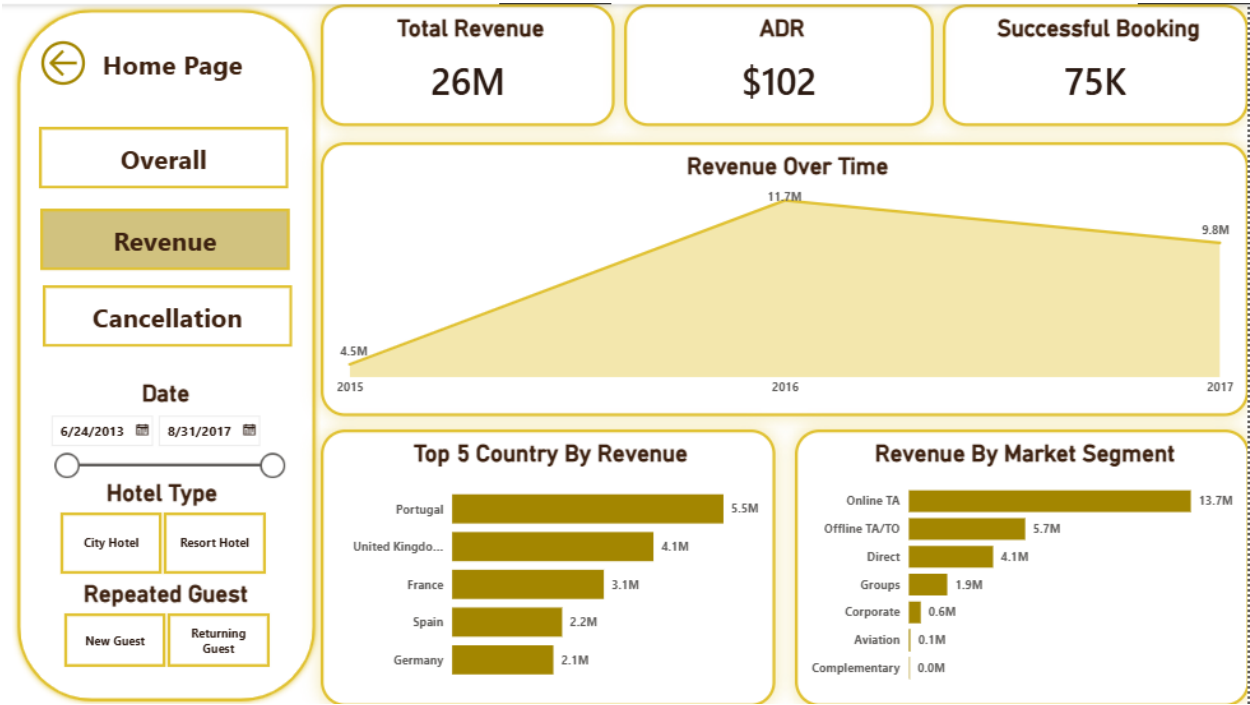
- Total Bookings
- Total Guests
- Cancellation Rate
- Average Stay Length
- Booking Trend
- Geographic Distribution of Bookings



Revenue Analysis Dashboard

Includes:

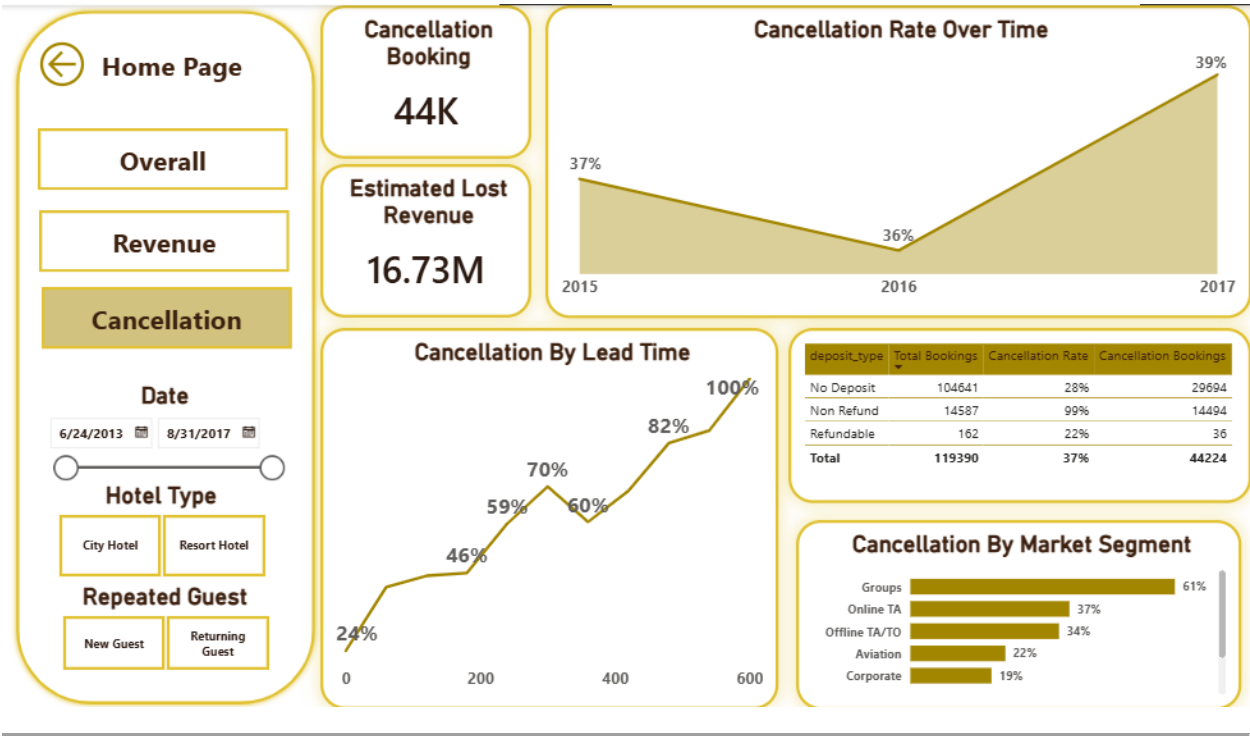
- Total Revenue
- ADR
- Successful Bookings
- Revenue Trend
- Revenue by Market Segment
- Top Revenue Generating Countries
- Revenue by Meal Type



Cancellation Analysis Dashboard

Includes:

- Cancellation Bookings
- Estimated Lost Revenue
- Cancellation Trend
- Cancellation Rate by Lead Time
- Cancellation by Deposit Type
- Cancellation by Market Segment



Key Business Insights

1. City Hotels Drive the Majority of Demand

Findings

- City Hotel bookings: approximately 79K
- Resort Hotel bookings: approximately 40K
- City Hotels hosted approximately 154K guests

- Resort Hotels hosted approximately 81K guests

Business Insight

City Hotels represent the primary source of booking demand and guest acquisition across the business.

2. Resort Hotel Guests Stay Longer

Findings

- Average stay length in City Hotels: 2.98 nights
- Average stay length in Resort Hotels: 4.32 nights

Business Insight

Resort Hotel customers tend to book longer stays, indicating that these properties primarily attract leisure and vacation travelers.

3. Cancellation Rates Are Significantly Higher in City Hotels

Findings

- City Hotel cancellation rate: 42%
- Resort Hotel cancellation rate: 28%

Business Insight

Bookings in City Hotels are substantially less stable and more likely to be canceled than bookings in Resort Hotels.

4. The Business Relies Primarily on New Guests

Findings

- New Guest bookings: approximately 116K
- Returning Guest bookings: approximately 4K

Business Insight

The business depends heavily on acquiring new customers, while repeat guest activity represents only a small portion of total bookings.

5. Returning Guests Are Much Less Likely to Cancel

Findings

- New Guest cancellation rate: 38%
- Returning Guest cancellation rate: 14%

Business Insight

Returning guests demonstrate significantly stronger booking commitment and are considerably less likely to cancel reservations.

6. Returning Guests Tend to Make Shorter Stays

Findings

- Average stay length for New Guests: 3.48 nights
- Average stay length for Returning Guests: 1.93 nights

Business Insight

Returning guests appear to make shorter and potentially more frequent visits compared to first-time guests.

7. Returning Resort Guests Represent the Most Stable Customer Segment

Findings

Returning Guests – City Hotel

- Bookings: approximately 2K
- Average stay length: 1.69 nights
- Cancellation rate: 22%

Returning Guests – Resort Hotel

- Bookings: approximately 2K
- Average stay length: 2.19 nights

- Cancellation rate: 6%

Business Insight

Returning guests staying at Resort Hotels represent the most reliable customer segment in the dataset.

8. Revenue Performance Overview

Findings

- Total realized revenue reached approximately \$26M.
- Revenue peaked during 2016 at approximately \$11.7M.
- Online Travel Agencies generated approximately \$13.7M in revenue.
- Portugal generated the highest revenue, followed by the United Kingdom, France, Spain, and Germany.

Business Insight

The business relies heavily on online distribution channels, while European markets represent the dominant revenue source.

9. Cancellation Risk Drivers

Findings

- Overall cancellation rate reached 37%.
- Estimated lost revenue reached approximately \$16.73M.
- Cancellation rates increased significantly as Lead Time increased.
- Group bookings recorded the highest cancellation rate among major market segments at 61%.
- Non-Refund bookings recorded a cancellation rate of approximately 99%.

Business Insight

Long-term reservations and group bookings represent the highest-risk booking categories and contribute disproportionately to cancellation exposure.

Final Conclusion

The analysis indicates that:

- City Hotels generate the highest booking volume but experience significantly higher cancellation rates.
- Resort Hotels attract fewer bookings but achieve longer stays and stronger booking stability.
- Revenue is concentrated within online booking channels and European markets.
- Returning guests consistently demonstrate higher booking reliability and lower cancellation behavior.
- Customer retention appears to be a key factor in reducing cancellations and improving business stability.
- Cancellation management represents a major opportunity for improving realized revenue and overall operational performance.